MIL-C-5191B (ASG)
20 MARCH 1953
Superseding
MIL-C-5191A(USAF)
19 March 1952

MILITARY SPECIFICATION

COMPUTER AND TRANSMITTER, TRUE AIRSPEED
PITOT STATIC, TYPE A-2

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of pitot static true airspeed computer and transmitter, designated Type A-2.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standard, drawings, and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-C-5015	Connectors; Electrical
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-D-5028	Drawings and Data Lists; Preparation of (for Engines, Accessories, and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads, Standard, Aeronautical

Air Force-Navy Aeronautical

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (for) Aluminum Alloys

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STANDARDS

MIL-STD-129	Marking of Shipments

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN507	Screw - Machine, Flat Head, 100 Degree
AN3102	Connector, Receptacle - Box Mounting
AND10401	Case - Standard Dimensions for 2-3/4 Dial Instrument (Without Sump)

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Protective treatment.- When materials are used in the construction of the computer and transmitter that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.1.2 Metals.- Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.1.3 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the computer and transmitter, except where magnetic materials are essential.

3.1.4 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.4.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.2 Design and construction.- The computer and transmitter shall conform to Drawing AND10401, except as shown on figure 1. The computer and transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.2.1 The computer and transmitter shall consist of a pitot static temperature-operated true airspeed computing mechanism operating a synchro control transformer whose electrical characteristics shall be as indicated in paragraph 3.2.2. The temperature sensing element shall conform to figure 2 and shall be supplied with necessary gaskets for flush mounting on the aircraft skin of pressurized cabin aircraft. Unless otherwise specified by the procuring activity, the length of the temperature capillary shall be 8 feet. The length of the capillary shall be marked on the case of the computer and transmitter.

3.2.2 Synchro-control transformer.- The synchro-control transformer shall be a one-phase, three-phase control transformer. The input impedance of the three-phase winding (line to line) shall be 25 ohms, or greater, at 400 cycles. When the three-phase winding is excited by 5.9V (line to line maximum) the output voltage per degree of error signal shall be at least 0.10V on open circuit. The resistance load across the output required to reduce open circuit output voltage to one-half its value shall be 3,800 ohms, or less, at 25°C.

3.2.2.1 The synchro shall be connected to the true airspeed mechanism so that the angular rotation of the synchro rotor shall be 36 degrees per 100 knots (true). Suitable stops shall be provided to prevent movement of the synchro rotor at speeds below 150 knots (true) or above 600 knots.

3.2.3 Case.- The computer and transmitter case shall consist of a body and cap and shall be in accordance with figure 1.

3.2.3.1 Body.- The body shall be made either of nonferrous low-density metal or of synthetic material, uniform in texture with a smooth surface, and entirely covered with a durable black finish. Synthetic material shall be of the thermal-setting type, composed of a suitable filler and phenol condensation product binder.

3.2.3.2 Reinforcement.- The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the computer and transmitter.

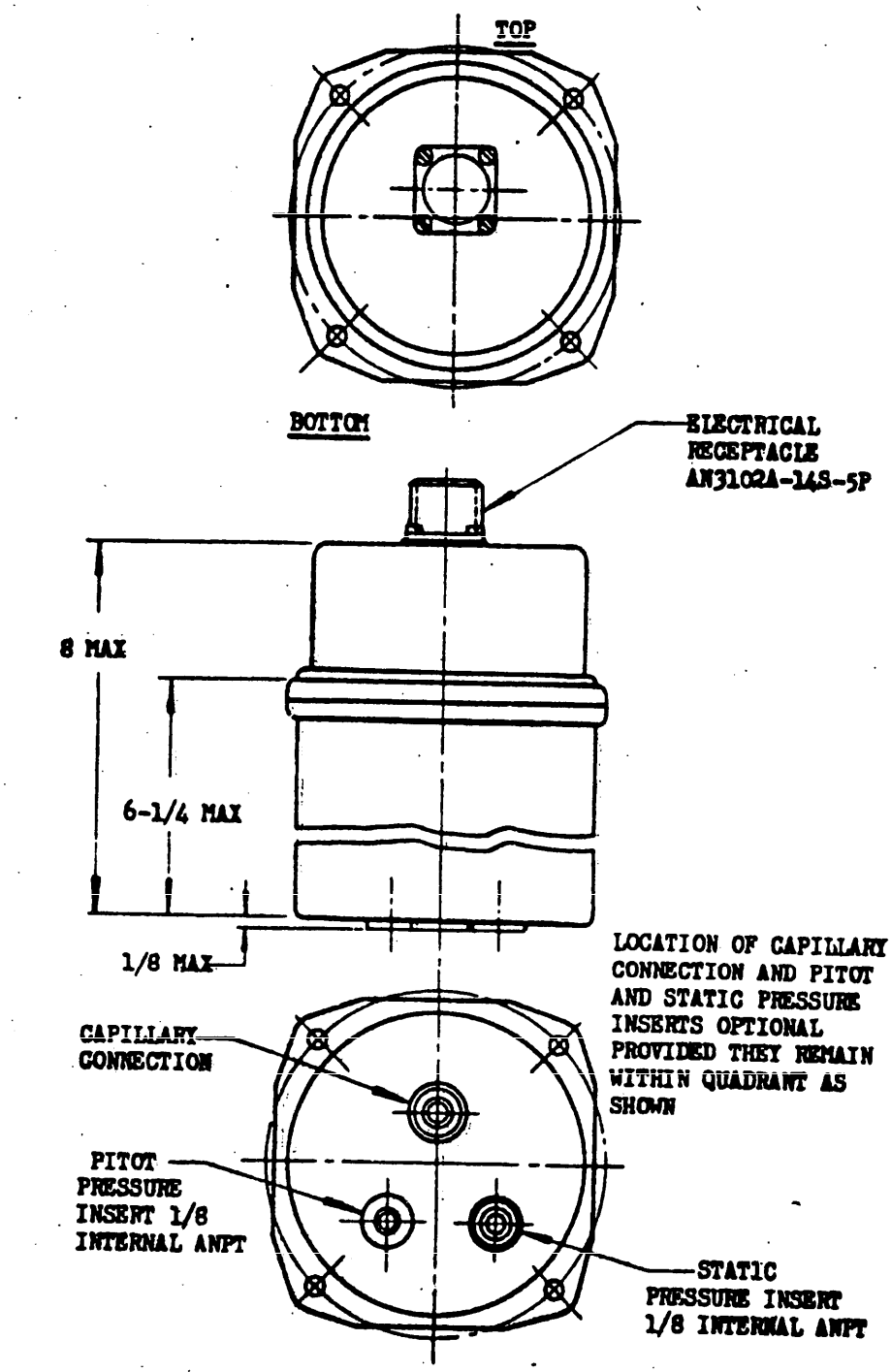
3.2.3.3 Capillary tubing.- The free air temperature capillary tubing shall be covered throughout its length with a flexible braided casing. The casing shall be securely attached to the case and to the bulb to protect the capillary tubing against strain.

3.2.3.4 Capillary reinforcement.- A suitable flexible reinforcement, extending 4 inches from the case and from the bulb, shall be provided on each end of the free air temperature capillary casing. The reinforcement shall be of a flexibility to withstand, without failure, a 180-degree bend of 1-3/8 inches radius. The reinforcement shall be securely attached to the case and to the temperature bulb. The inside diameter of the reinforcement shall be such that the reinforcement will slide freely on the capillary casing. The reinforcement shall be braided brass or bronze spiral armor, approximately 9/32 inch outside diameter, plain weave, seven strands. The brass or bronze wires shall be 0.011 inch in diameter with 6-1/2 to 7 strands per inch.

3.2.3.5 Case marking.- Except as specified herein, the exterior of the case shall be finished in durable dull black.

3.2.3.5.1 The back of the case, adjacent to the pressure connections, shall be marked as follows:

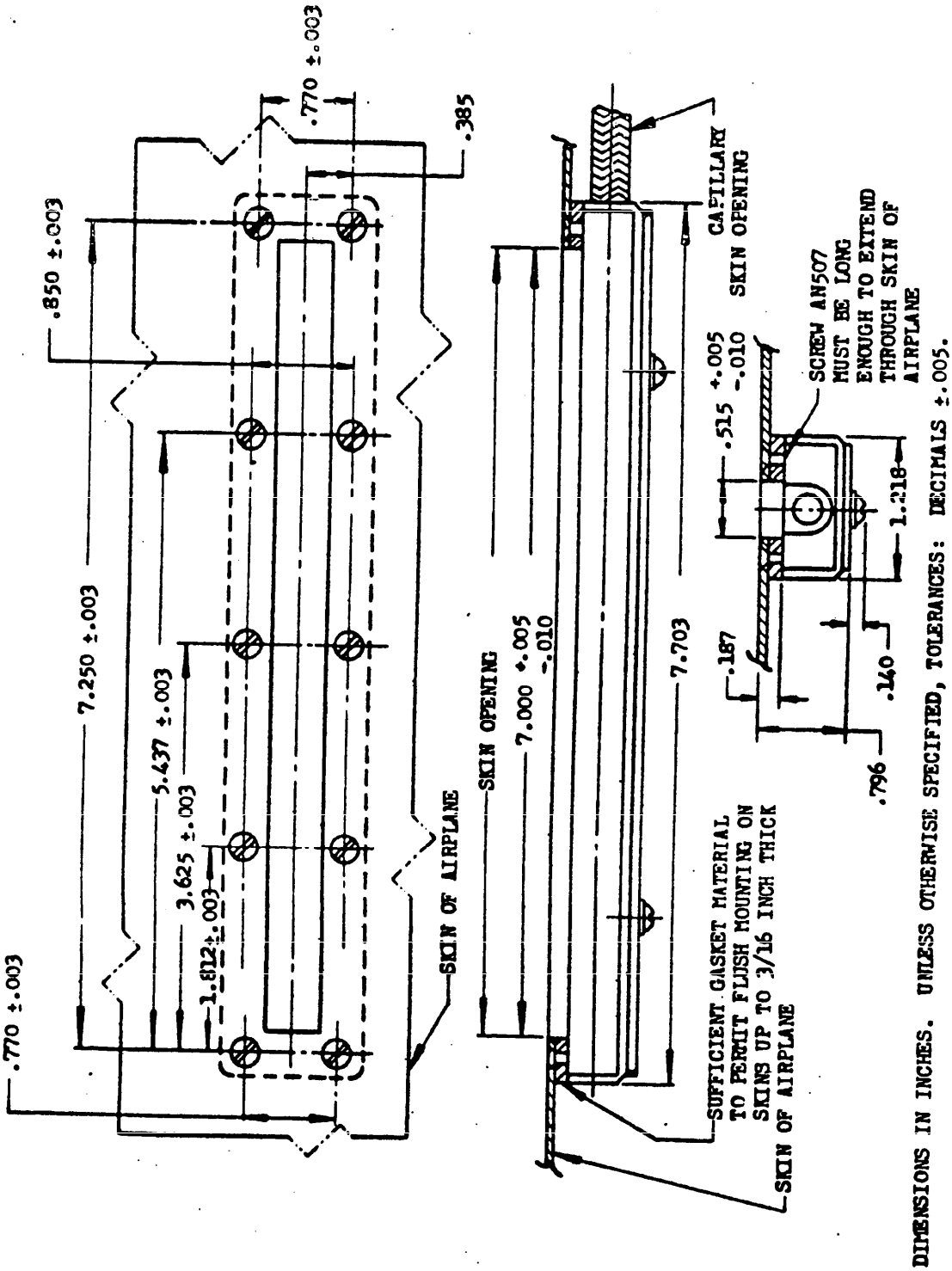
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DIMENSIONS IN INCHES.

FIGURE 1. Case - computer and transmitter

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.005.

FIGURE 2. Temperature sensing element

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"P" Pressure connection
"S" Static connection

3.2.3.5.2 The following shall be marked on the back of the case:

CAUTION
DO NOT BLOW IN TUBES

3.2.4 Electrical connector.- The electrical connector used in the construction of the computer and transmitter shall be in accordance with Specification MIL-C-5015.

3.2.4.1 The pins of the electrical connector shall be connected to the synchro as follows:

Pin A - Single-phase winding No. 1
Pin B - Single-phase winding No. 2
Pin C - Three-phase winding No. 1
Pin D - Three-phase winding No. 2
Pin E - Three-phase winding No. 3

The phase rotation shall be C-D-E for airspeed increasing.

3.2.5 Electrical zero.- The electrical zero of the computer and transmitter shall be set at 300 knots true airspeed at an altitude of 15,000 feet and a bulb temperature of 0°C.

3.2.6 The indicator shall be so designed that proper compensation is made for variation in ambient temperature.

3.3 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.4 Application of pressure.- The pitot pressure shall be applied to the inside of the diaphragm capsule, and the static pressure shall be applied to the interior of the case.

3.5 Threads.-

3.5.1 Screw threads.- Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.5.2 Pipe threads.- Pipe threads shall conform to Specification MIL-P-7105.

3.6 Weight.- The weight of the computer and transmitter completely assembled, less mounting nuts and mounting screws, shall not exceed 4 pounds.

3.7 Finishes and protective coatings.-

3.7.1 Aluminum-alloy parts.- Aluminum-alloy parts shall be covered with an anodic film conforming to Specification AN-QQ-A-696, except as follows.

3.7.1.1 Small holes and case inserts need not be anodized.

3.7.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.7.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with Specification MIL-C-5541 may be used in lieu of anodizing.

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3.7.1.4 Castings containing non-aluminum-alloy integral inserts may be treated with a chemical film in accordance with Specification MIL-C-5541 in lieu of anodizing.

3.7.1.5 When abrasion resistance is a factor, chemical films in accordance with Specification MIL-C-5541 shall not be used in lieu of anodizing.

3.7.2 Steel parts.- Steel parts shall be cadmium-plated, where practicable, in accordance with Specification QQ-P-416, type II or III, and of a class that is adequate to achieve the degree of protection required.

3.8 Identification of product.-

3.8.1 Nameplate.- The computor and transmitter shall be marked for identification in accordance with Specification MIL-M-7911. The nameplate shall contain the following information:

COMPUTER AND TRANSMITTER, TRUE AIRSPEED, 150-600 KNOTS
Type A-2
Specification MIL-C-5191B(ASG)
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
US Property

3.9 Installation.-

3.9.1 Mounting screws.- The contractor shall furnish sufficient mounting screws for installing the computer and transmitter. They shall be No. 6-32 NC-2 round head brass machine screws having a durable dull black finish. Length shall be sufficient to permit mounting on a panel 3/16 inch thick.

3.9.2 Envelope.- An envelope, containing the mounting screws, shall be furnished with each computer and transmitter. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.9.3 Accessories.- The contractor shall furnish with each computer and transmitter a sun shield, gaskets, and sufficient mounting screws and nuts for installation of the temperature bulb.

3.10 Workmanship.-

3.10.1 General.- The computer and transmitter, including all parts and accessories, shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.10.2 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the computer and transmitter, they shall be held or limited accordingly.

3.10.3 Screw assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

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3.10.4 Cleaning.- The computer and transmitter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of computers and transmitters shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on computers and transmitters manufactured and submitted for acceptance under contract.

4.2 Test conditions.-

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Tapping.- Unless otherwise specified, the computer and transmitter shall be gently tapped or vibrated before a test reading is taken.

4.2.3 Attitude.- Unless otherwise specified, the computer and transmitter shall be tested in normal operating position.

4.2.4 Vibration.- A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cps and shall subject the computer and transmitter to vibration such that a point on the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.2.5 Unless otherwise specified, whenever indications from the computer and transmitter are required, the synchro control transformer shall be connected to a follow-up synchro through an amplifier and servo system, as shown on figure 3. Facilities for determining the rotor position of the follow-up synchro to within ±0.1 degree shall be provided and the servo system shall not introduce more than 0.1 degree error in the indication. The single-phase winding of the follow-up synchro shall be excited with 26V 400 cps, and the error signal for the amplifier shall be obtained from the single-phase winding of the synchro on the computer and transmitter. The indicator shall be set to indicate 72 degrees when connected to the computer and transmitter at the electrical zero.

4.3 Qualification tests.-

4.3.1 Prior qualification.- Unless otherwise specified by the procuring activity, computers and transmitters which have not previously passed a Qualification test, or which have passed the Qualification tests and have been modified in any manner, shall satisfactorily pass a Qualification test prior to the acceptance of any computer and transmitter.

4.3.2 Sampling instructions.- Qualification test samples shall consist of three computers and transmitters. Samples shall be identified as required and forwarded to

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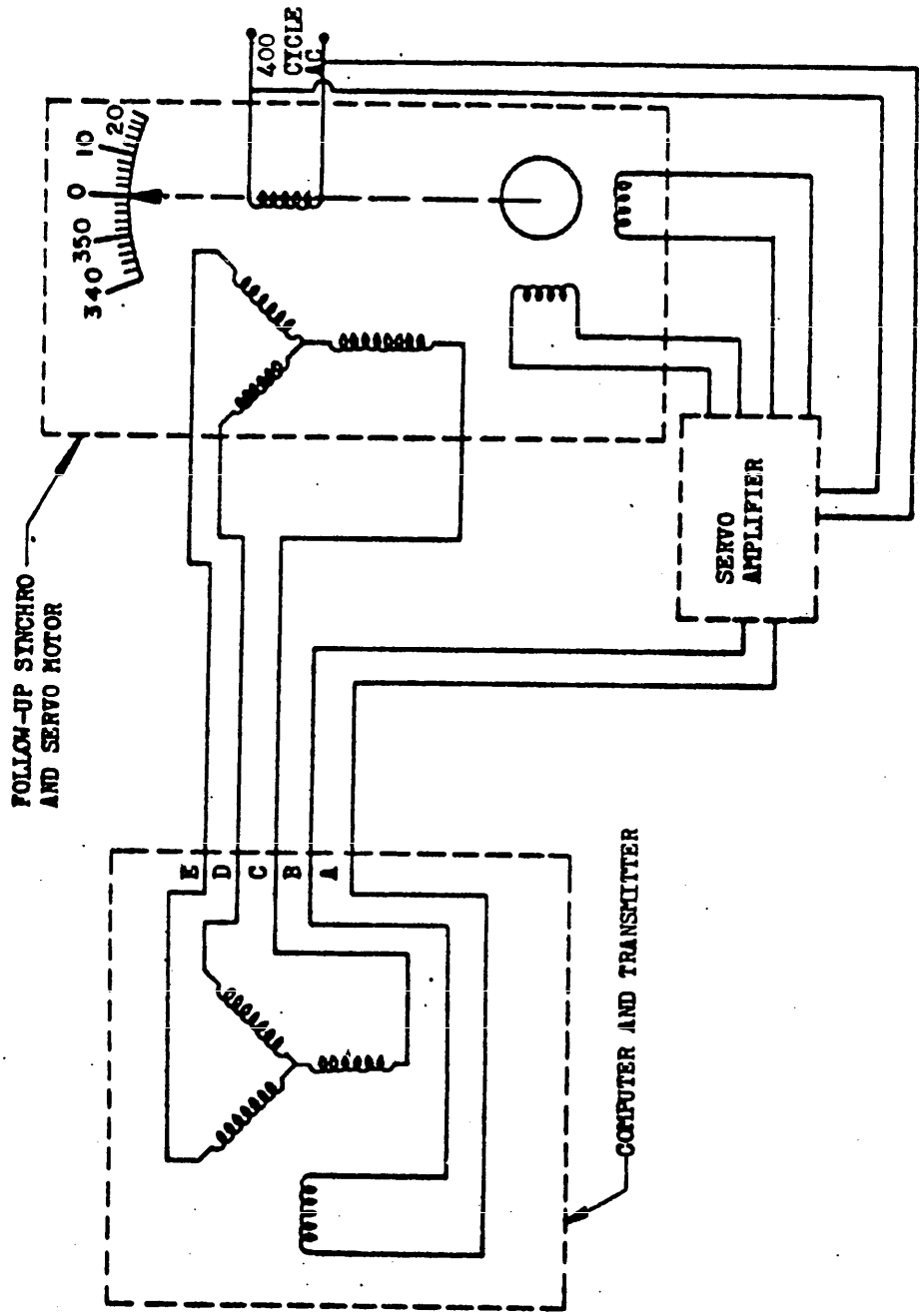


FIGURE 3. Wiring diagram - follow-up synchro

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the activity responsible for qualification, designated in the letter of authorization from the qualifying activity. (See paragraph 6.3.)

4.3.3 Tests.- The Qualification tests shall consist of the Inspection tests and the following tests.

4.3.3.1 Environmental.- The following tests shall be in accordance with the specified procedure of Specification MIL-E-5272, and as specified herein.

4.3.3.1.1 Low temperature: Procedure II.- Before completion of the 24-hour period, pressure shall be applied to the pitot connector and electrical power to the servo system. The computer and transmitter shall operate without sticking. After the computer and transmitter has returned to room temperature, it shall meet the requirements specified for room temperature scale error. There shall be no damage which would adversely affect subsequent operation.

4.3.3.1.2 High temperature: Procedure I, for a period of 24 hours.- After the computer and transmitter has returned to room temperature, it shall meet the requirements specified for room temperature scale error.

4.3.3.1.3 Altitude: Procedure II, - except that the altitude shall be 60,000 feet instead of 50,000 feet at an ambient temperature of -54°C and a dewpoint of -57° to -70°C for a continuous period of 4 hours. The computer and transmitter shall be capable of satisfactory operation and no visible evidence of physical damage upon completion of this test.

4.3.3.1.4 Vibration: Procedure V.- The test shall be conducted while a differential pressure of 400 knots is applied to the computer and transmitter. The difference between the indication of the pointer at the beginning and at the end of the 3-hour vibration period shall not exceed the tolerance specified in table I, and no looseness in the mechanism nor damage shall result in any part of the computer and transmitter.

TABLE I

Pointer vibration test tolerances

Tests	Tolerances ±
Friction	2.0 degrees
Position error	2.0 degrees
Vibration error	
Null signal increase	0.2V
Null shift tolerance	2.0 degrees
Vibration failure	2.0 degrees
Scale error	
Low temperature operation	2.0 degrees
High temperature operation	2.0 degrees
Seasoning	2.0 degrees
Overpressure	2.0 degrees
Capillary temperature	1.0 degrees
Capillary bending and strain	1.0 degrees

4.3.3.1.5 Humidity: Procedure I.- The computer and transmitter, including applicable external connections, shall be mounted in its normal operating position. There shall be no evidence of corrosion or rusting of the internal or external parts of the computer and transmitter.

4.3.3.1.6 Fungus: Procedure I.- The computer and transmitter shall have applicable external connections installed. After the test, the computer and transmitter shall meet the requirements specified for Individual tests. There shall be no deterioration which would adversely affect subsequent operation.

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4.3.3.2 Acceleration.- The computer and transmitter shall be mounted and accelerated first in its vertical axis and then in each of two axes that are perpendicular to the vertical axis and to each other. The computer and transmitter shall be subjected to an acceleration of 20g in each of the above axes for a period of 30 seconds. At the end of this acceleration, the computer and transmitter shall meet the requirements specified for room temperature scale error. No damage to the computer and transmitter shall result from this test.

4.3.3.3 Mounting lugs.- The computer and transmitter case with the mechanism and cap removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for 1 minute to each lug along the lug axis and toward the rear of the instrument. The lugs shall withstand the applied load without fracture.

4.4 Inspection tests.- The Inspection tests shall consist of Individual tests and Sampling tests.

4.4.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.4.2 Individual tests.- Each computer and transmitter shall be subjected to the following tests.

4.4.2.1 Examination of product.- The computer and transmitter shall be inspected to determine compliance with this specification with respect to material, workmanship, and envelope.

4.4.2.2 Dielectric strength.- A potential of 500V ac at commercial frequency shall be applied between each pin and the case of the computer and transmitter for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.4.2.3 Electric zero.- The computer and transmitter shall be tested for electrical zero as follows.

4.4.2.3.1 Leads from pin C, E, and A shall be tied together. A 0.1-uf condenser shall be connected across leads from pins A and B. A voltmeter shall be connected between leads for pins B and D. A voltage of 9V 400 cycles shall be applied between C and D. With the connections specified in paragraph 4.4.2.4, the bulb temperature shall be adjusted to 0°C and the pressure altitude to 15,000 feet. The differential pressure (indicated airspeed) shall be slowly increased. The voltage should increase as the indicated airspeed is increased from 150 to 200 knots. If the voltage increases, the voltmeter lead shall be disconnected from pin D and connected to pin A. The indicated airspeed shall be increased until the voltmeter indicates the minimum voltage. This is the electrical zero. The indicated airspeed at the minimum voltage shall be 238 ±2 knots.

CAUTION: DO NOT EXCEED 350 KNOTS INDICATED AIRSPEED IN THIS TEST.

4.4.2.4 Scale error.- The computer and transmitter shall be mounted on a panel that has facilities for tapping to remove the friction effect. The free air temperature bulb shall be placed in a free air thermometer test fixture in which it is possible to control, to vary, and to read the temperature accurately throughout the range of from -60° to +40°C.

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4.4.2.4.1 The static pressure connection of the computer and transmitter shall be connected by means of a cross to an accurate barometer, a source of air pressure and suction, and to the suction side of a manometer.

4.4.2.4.2 The pitot pressure connection of the computer and transmitter shall be connected by means of a tee to a source of air pressure and suction and to the pressure side of the manometer.

4.4.2.4.3 The electrical connection shall be as specified in paragraph 4.2.5.

4.4.2.4.4 Each computer and transmitter shall be tested for scale errors as follows:

(a) The temperature around the free air temperature bulb in the test fixture shall be adjusted to the temperature specified in column 1 of table II.

(b) The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude, sea level standard, as specified in column 2 of table II.

(c) The pitot pressure shall then be adjusted so that the differential pressure indicated by the manometer is equal to the first test setting, as specified in column 3 of table II.

(d) While the temperature of the free air temperature bulb is maintained at the first test temperature setting, as specified in table II, and the static pressure adjusted to zero feet altitude, the pitot pressure shall be varied so that the differential pressure is equal to each of the values marked by an asterisk listed in column 3 of table II. Under these conditions, the follow-up synchro indicator shall indicate the angular displacement listed in column 5 and shall not vary from those indications by more than the tolerances listed in column 6. This procedure shall be followed for each of the different pressure altitudes listed in table II.

(1) The procedure described in paragraph 4.4.2.4.4 (d) shall be repeated for the conditions specified in tables III, IV, V, VI, and VII.

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TABLE II

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
+40	0	150	156	20	4
		200 *	207	39	3
		250	258	57	3
		300 *	308	75	3
		350 *	357	93	3
		400 *	405	110	3
		450	453	127	4
		500 *	499	144	4
		550	544	160	5
		600 *	588	176	5
	5,000	150	175	25	4
		200 *	226	45	3
		250	281	65	3
		300 *	335	85	3
		350	387	103	3
		400 *	438	122	3
		450	488	140	4
		500 *	536	157	5
		550	583	174	5

* Test points

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TABLE III

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
+20	0	150	151	18	4
		200 *	200	36	3
		250	249	54	3
		300 *	298	71	3
		350	345	88	3
		400 *	392	105	3
		450	438	122	3
		500 *	483	138	4
		550	526	153	5
		600 *	569	169	5
	5,000	150	165	23	4
		200 *	219	43	3
		250	272	62	3
		300 *	324	81	3
		350	375	99	3
		400 *	424	117	3
		450	472	134	4
		500 *	519	151	4
		550	564	167	5
	10,000	150	181	29	4
		200 *	239	50	3
		250	297	71	3
		300 *	353	91	3
		350	407	111	3
		400 *	459	129	4
		450	510	147	4
		500 *	558	165	5

* Test points

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TABLE IV

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
0	0	200 *	193	33	4
		250	241	51	3
		300 *	287	67	3
		350	333	84	3
		400 *	379	100	3
		450	423	116	3
		500 *	466	132	4
		550	508	147	4
		600 *	549	162	5
		650	589	176	5
	5,000	150	159	21	4
		200 *	211	40	3
		250	262	58	3
		300 *	312	76	3
		350	362	94	3
		400 *	410	112	3
		450	456	128	4
		500 *	501	144	4
		550	545	160	5
		600 *	587	175	5
	10,000	150	175	27	4
		200 *	211	40	3
		250	262	58	3
		300 *	312	76	3
		350	362	94	3
		400 *	410	112	3
		450	492	141	4
		500 *	539	158	5
		550	583	174	5
	15,000	150	192	33	4
		200 *	254	55	3
		250	314	77	3
		300 *	372	98	3
		350	427	118	3
		400 *	480	137	4
		450	531	155	4
		500 *	579	172	5

* Test points

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TABLE V

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
-20	0	200 *	186	31	4
		250	232	48	3
		300 *	277	64	3
		350	321	80	3
		400 *	364	95	3
		450	407	111	3
		500 *	443	125	3
		550	489	140	4
		600 *	528	154	4
		650	567	168	5
	10,000	150	168	24	4
		200 *	223	44	3
		250	276	63	3
		300 *	328	82	3
		350	378	100	3
		400 *	427	118	3
		450	474	135	4
		500 *	519	151	4
		550	562	166	5
	15,000	150	185	31	4
		200 *	244	52	3
		250	302	73	3
		300 *	358	93	3
		350	411	112	3
		400 *	462	130	4
		450	511	148	4
		500 *	557	165	5
	20,000	150	204	37	4
		200 *	269	61	3
		250	331	83	3
		300 *	391	105	3
		350	447	125	3
		400 *	501	144	4
		450	551	162	5

* Test points

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TABLE VI

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed Knots	Indication degrees	Tolerance degrees
-40	0	200 *	179	28	4
		250	222	44	3
		300 *	265	59	3
		350	308	75	3
		400 *	350	90	3
		450	390	104	3
		500 *	430	119	3
		550	469	113	4
		600 *	507	147	4
	10,000	150	161	22	4
		200 *	213	41	3
		250	265	59	3
		300 *	315	77	3
		350	363	95	3
		400 *	409	111	3
		450	454	127	4
		500 *	498	143	4
	20,000	150	196	35	4
		200 *	258	57	3
		250	318	78	3
		300 *	375	99	3
		350	429	118	3
		400 *	480	137	4
		450	529	154	4
	35,000	150	268	60	3
		200 *	349	90	3
		250	423	116	3
		300 *	491	141	4
	50,000	150	369	97	3
		200 *	468	132	3

* Test points

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TABLE VII

Scale error test - settings and tolerances

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
	0	200 *	171	26	4
		250	213	41	3
		300 *	254	55	3
		350	294	70	3
	10,000	150	154	19	4
		200 *	204	37	3
		250	253	55	3
		300 *	301	72	3
	20,000	150 *	187	34	4
		200	247	53	3
		250 *	304	73	3
	35,000	150 *	256	56	3

* Test points

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4.4.2.4.4.1 True airspeed knots in the tables of this specification are calculated from the following equations:

$$\text{Indication} = 36\ (V_{true} - 150)$$

$$V_{true} = 87.0114\ T_o \sqrt{\left[\left(1 + \frac{dP}{P}\right)^{0.2857} - 1\right]} \qquad (1)$$

Where V_{true} = true airspeed in knots
T_o = corrected temperature in degrees K (Free air temperature)

$$T_o = \left(273.2 + T\right) - \left(1.07\ (V_{true})^2 \times 10^{-4}\right) \qquad (2)$$

Where T is test temperature in degrees Centigrade
P = pressure corresponding to given altitude
dP = pressure corresponding to given indicated airspeed (same units as P)

4.4.2.4.5 The Scale error tests shall be made by subjecting the computer and transmitter to the differential pressures specified to produce the readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to give the desired reading; and with pressures decreasing, the pressure shall be brought down to, but shall not fall below the pressure specified, to give the desired reading. The errors at the test points shall not exceed the tolerances specified in tables II to VII, inclusive.

4.4.2.4.6 When desired by the Inspector, the computer and transmitter may be tested for scale error at any additional point or points on the scale, in combination with any indicated airspeed listed in the tables, and any pressure altitude and free air temperature. The errors at these additional test points shall not exceed the tolerance listed in the table.

4.4.2.5 Friction.- The computer and transmitter shall be tested for friction at every point indicated by an asterisk in tables II to VII, inclusive. The pressure shall be increased, to bring the pointer approximately to the desired reading, and then held constant while two readings are taken; the first before the computer and transmitter is tapped, and the second after the computer and transmitter is tapped. The difference between any two such readings shall not exceed the tolerance specified in table I. The pointer shall move smoothly, while the pressure is varied uniformly, with vibration of the computer and transmitter. This test may be combined with the test for scale errors. During this test the servo indicator may be tapped to remove any friction in the indicator and servo system.

4.4.2.6 Leakage.-

4.4.2.6.1 Case leak.- The pitot and static pressure connections of the computer and transmitter shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be applied to the pitot and static pressure connections of the computer and transmitter. With the source disconnected during a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

4.4.2.6.2 Diaphragm capsule leak.- The static connection shall be opened to the atmosphere and the pitot connection shall be connected to a mercury manometer and to a source of pressure. A pressure of 20 inches Hg shall slowly be applied to the pitot pressure connection of the computer and transmitter. With the source of pressure sealed during a period of 1 minute, the difference between the mercury levels in the manometer shall not change by more than 0.1 inch.

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4.4.3 Sampling tests.- Five computers and transmitters shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following Sampling tests. A lot shall consist of transmitters manufactured under essentially the same conditions, submitted for inspection at substantially the same time, and shall not exceed 500 units. These tests shall be in addition to the Individual tests.

4.4.3.1 Weight.- The computer and transmitter shall be weighed completely assembled, except for connecting fittings, sun shield, and mounting screws.

4.4.3.2 Magnetic property.- The magnetic property of the computer and transmitter shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.180 ±0.001 gauss when the computer and transmitter are held in various positions on an east-west line 5 inches from the center of the magnet. An aircraft compass with compensating magnets removed therefrom may be used as the magnet for this test. The maximum deflection of the magnet shall not exceed 2 degrees. This test shall be conducted with electrical power connected to the synchro.

4.4.3.3 Low temperature operation.- The computer and transmitter shall be placed in a chamber and maintained at a temperature of -54°C for a period of 4 hours with no electrical power applied. While the computer and transmitter is still at this temperature, power shall be applied; and the computer and transmitter shall meet the requirements specified for scale errors as specified in paragraph 4.4.2.4, except that only tables VI and VII shall be used. The errors at the test points shall not exceed the tolerances specified in tables VI and VII by more than the tolerances specified in table I.

4.4.3.4 High temperature operation.- The computer and transmitter shall be placed in a chamber and maintained at a temperature of +71°C for a period of 4 hours with no electrical power applied. While the computer and transmitter is still at this temperature, power shall be applied; and the computer and transmitter shall meet the requirements specified for scale error as specified in paragraph 4.4.2.4, except that only tables II and III shall be used. The errors at the test points shall not exceed the tolerances specified in tables II and III by more than the tolerances specified in table I.

4.4.3.5 Seasoning.- The pitot pressure connection of the computer and transmitter shall be subjected to 1,000 applications of a differential pressure of approximately 20 inches Hg. These applications shall be made at a rate not greater than 6 per minute. Not less than 1 hour following this test, the computer and transmitter shall be tested for scale errors as specified in paragraph 4.4.2.4, at +20°C and 5,000 feet pressure altitude. The errors at the test points shall not exceed the tolerance specified in table V by more than the tolerances specified in table I.

4.4.3.6 Capillary temperature.- A 3-foot length of capillary shall be subjected to a temperature of 50°C, with the computer and transmitter, the bulb, and the remainder of the capillary at room temperature. The change in reading, due to heating the capillary shall not exceed the tolerances given in table I.

4.4.3.7 Capillary bending and strain.- The computer and transmitter indicator reading shall be obtained for a bulb temperature of 50°C. The capillary shall be bent 20 times in one place on a radius of 1/2 inch through an angle of 90 degrees and straightened out after each bending. The flexible braided capillary covering, at points approximately 6 inches from the case and from the bulb, shall be secured in a vise so that the case and the bulb are suspended vertically downward. Care shall be taken to secure the flexible covering lightly in the vise to avoid crushing the covering. The change in reading as the result of bending and straining the capillary shall not exceed the tolerances given in table I.

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4.4.3.8 Vibration error.- The computer and transmitter shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane and with the plane of the mounting lugs of the computer and transmitter vertical. This test shall be conducted while differential pressures of 200 and 400 knots are applied to the computer and transmitter. Electrical connections shall be made to the computer and transmitter as shown on figure 4 and the standard synchro rotated to the null voltage as shown by the lowest amplitude signal on the oscilloscope. The computer and transmitter shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm, in order to determine that the natural frequency of the computer and transmitter does not occur in this frequency range. While the computer and transmitter is being vibrated, the maximum amplitude of the null signal shall not increase in value more than the amount specified in table I. The angular shift of the standard synchro to maintain the null shall be no greater than the amount specified in table I.

4.4.3.9 Overpressure.-

4.4.3.9.1 A pressure of 30 inches Hg shall be applied to the pitot pressure connection of the computer and transmitter for a period of 10 minutes.

4.4.3.9.2 A pressure of 6 inches of water shall be applied to the static connection of the computer and transmitter for a period of 10 minutes.

4.4.3.9.3 Not less than 5 minutes following the application of the Overpressure tests, the computer and transmitter shall be tested for scale error as specified in paragraph 4.4.2.4 at +20°C and 5,000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in table III by more than the tolerances specified in table I.

4.4.3.10 Position error.- The indicator shall be observed while the computer and transmitter is held in any desired position and tapped. This reading shall not differ from the reading, when the computer and transmitter is held in any other position, by more than the tolerances specified in table I. This test shall be made at every other one of the points indicated by an asterisk in the tables beginning with table II.

4.4.4 Rejection and retest.- Individual units failing to meet their respective tests shall be rejected. When tests are specified on a quantity of units that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, the lot shall be held in abeyance until the extent and cause of failure is determined. Individual performance tests need not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the equipment, in which case the lot shall be rejected. Rejected units shall be replaced or shall be reworked to correct the defects, after which all necessary tests shall be repeated. Rejected equipment shall not be resubmitted for inspection without furnishing all particulars concerning previous rejections and measures taken to overcome the defects.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.- Each computer and transmitter shall be packaged and packed for shipment in accordance with Specification MIL-P-5633.

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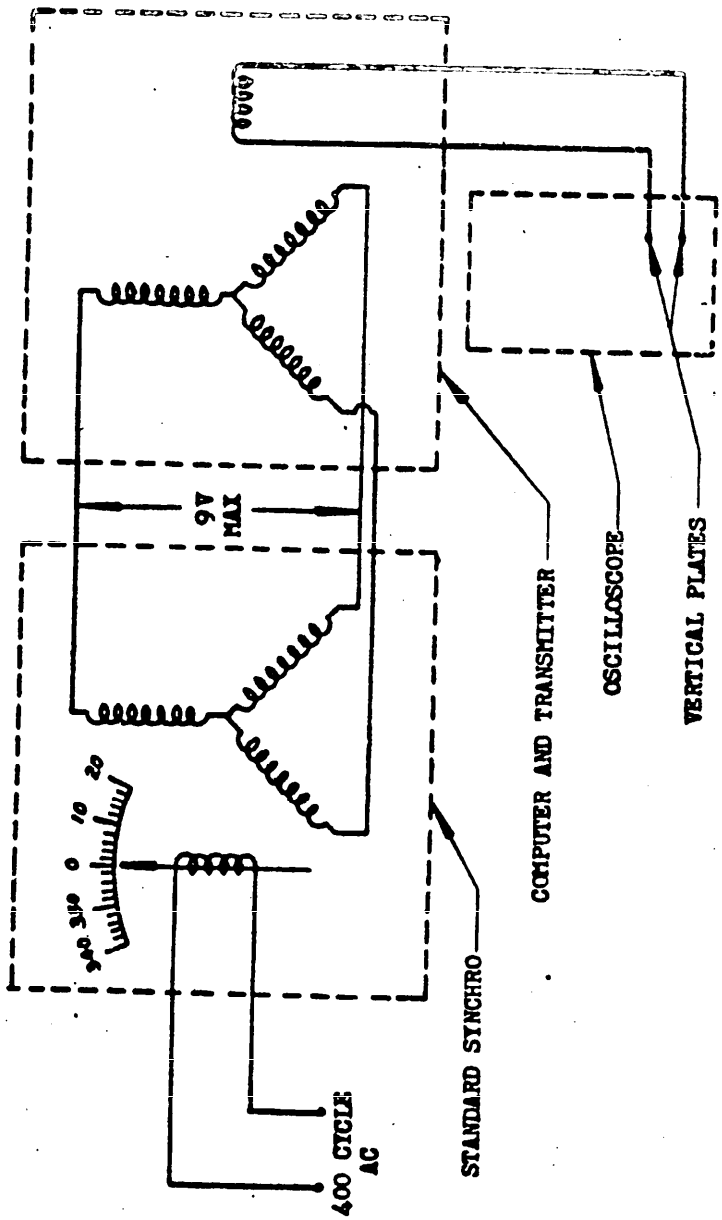


FIGURE 4. Electrical connections for Vibration error test

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5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Computer and Transmitter, True Airspeed, Pitot Static 150-600 Knots, Type A-2, (*Manufacturer's Part No.), Specification MIL-C-5191B(ASG).

*Applicable data to be entered by the contractor.

5.3.1 The following information shall be included, for the Air Force only.

IF IN STORAGE AFTER (DATE) 1/, THE COMPUTER AND TRANSMITTER SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

1/ This date shall be 36 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

6. NOTES

6.1 Intended use.- The Type A-2 computer and transmitter covered by this specification is intended for use on aircraft to supply signals for indication of true airspeed and for operating various navigation and armament equipment requiring inputs of true airspeed. The true airspeed range is from 150 to 600 knots at altitudes of 0 to 50,000 feet.

6.2 Definitions.-

6.2.1 Amplitude.- Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.2.2 External connections.- When external connections to simulate installed conditions are called for, they shall consist of 10 feet of 1/4-inch copper or aluminum tubing coiled so that moisture can drain out the open end. Similar coils shall be connected to the pitot and static pressure connections.

6.3 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, the activity responsible for qualification, with a copy to the Bureau of Aeronautics, Navy Department, Washington 25, D.C., and arrange to have the products that they propose to offer to the Navy or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the above designated activity responsible for qualification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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